Class Project Proposal
Semantic Segmentation with Dilated CNNs for Autonomous Driving

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Project Idea

For this class project, I want to study semantic segmentation for autonomous driving. Semantic segmentation means assigning a label to every pixel in an image, such as road, sidewalk, car, pedestrian, sky, or traffic sign. I am interested in this topic because it connects computer vision to a real application that people use and care about. In autonomous driving, understanding the scene correctly is important for safety, navigation, and decision-making.

My main goal is to build a simple semantic segmentation pipeline and then improve it using dilated convolutions. I want to compare a basic CNN with a dilated CNN and study whether dilation helps preserve more spatial detail while still keeping the model practical for class project work.

Reason behind choosing this project is that it is both interesting and useful. It builds on ideas from image segmentation and convolutional neural networks, but it also has a clear real-world civil engineering purpose: answering this question of how a model can look at a road scene and separate important classes such as roads, cars, and pedestrians.

Main Question

The main question of this project is simple: can a dilated CNN improve semantic segmentation for driving scenes compared to a basic CNN, while still remaining manageable in terms of computation and implementation?

The goal is not to test every advanced model, but to understand the problem clearly, build a reasonable baseline, and then explore whether dilated convolutions provide a meaningful improvement.

Background

In standard CNNs, pooling and subsampling help reduce computation, but they also reduce spatial resolution. For semantic segmentation, that can be a problem because the model needs detailed pixel-level information. Dilated convolutions are useful here because they enlarge the receptive field without shrinking the feature map as much. That makes them a natural idea to explore for dense prediction tasks [1, 2].

Earlier work such as Fully Convolutional Networks showed that CNNs could be used for pixel-wise prediction [4]. Later work on dilated convolutions showed that it is possible to improve context aggregation while keeping more spatial information [1]. These ideas motivate the direction of this project.

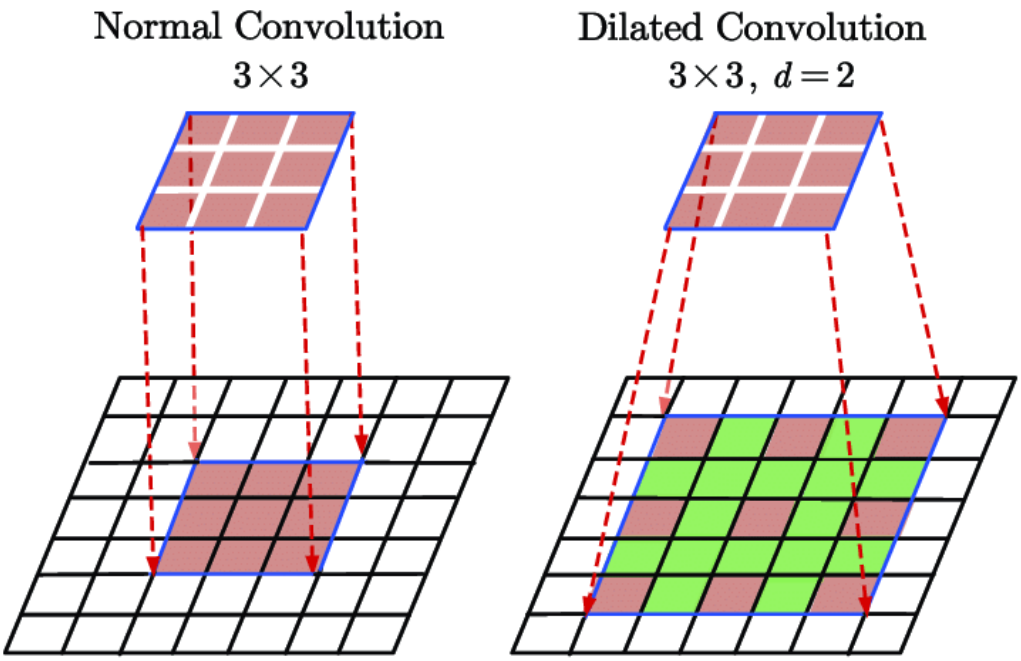

Figure 1: Dilated convolution increases the receptive field without reducing spatial resolution. Figure obtained from [3].

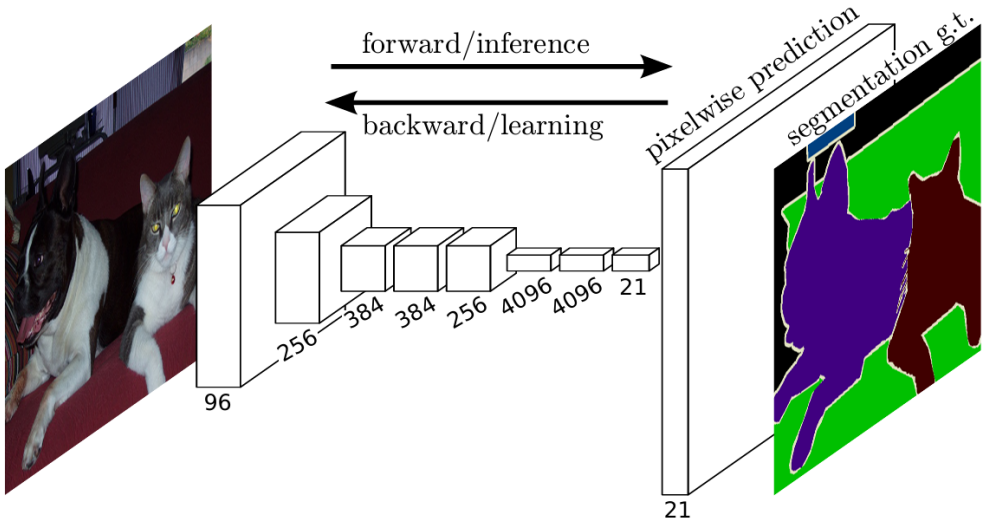

Figure 2: Fully convolutional networks helped establish semantic segmentation as a dense prediction problem. Figure obtained from [4].

Dataset

It is planed to use the Cityscapes dataset because it is a well-known dataset for semantic segmentation in urban driving scenes. It includes street images and fine annotations for important classes such as road, sidewalk, vehicles, and pedestrians. It is a strong fit for the project because it directly matches the autonomous driving setting. In Figure 3, the annotated image illustrates that each

pixel is assigned a corresponding label; for example, humans are shown in red, sidewalks in pink, and trains in turquoise.

Figure 3: Example of a fine-annotated image from the Cityscapes dataset.

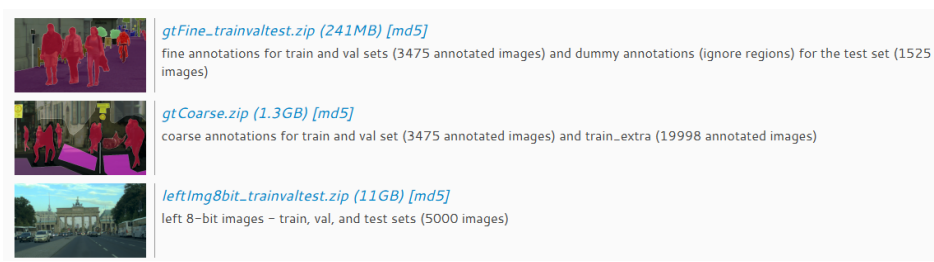

Figure 4: Download page of the Cityscapes dataset: Link

I will likely use the fine-annotated labels together with the leftImg8bit image dataset. Of course, both will need to be resized to a lower resolution so that training is feasible on the available hardware. This will keep the project manageable while still allowing for meaningful experimentation.

Planned Approach

My plan is to complete the project in two main steps.

- First, I will build a simple baseline CNN for semantic segmentation so I have a clear starting point.
- Second, I will modify that model by introducing dilated convolutions and compare the two approaches.

I want to keep the implementation understandable rather than overly complex. If I achieve the proposed results, I may be able to go beyond this initial understanding, implement the core idea, and further expand the project along the way. I may extend it by exploring more advanced

architectures or using a pre-trained network to compare the results. The project is mainly focused on understanding how dilated convolutions affect the behavior of a CNN, and whether that change improves scene understanding in road images.

Evaluation Plan

I plan to evaluate the models using mean Intersection over Union (mIoU), since it is a common metric for semantic segmentation. I also want to look at prediction images directly so I can visually compare how well each model separates roads, cars, pedestrians, and other classes. It is defined as the average of the Intersection over Union (IoU) across all K classes defined as follows

$$\mathrm{IoU}_k = \frac{TP_k}{TP_k + FP_k + FN_k}$$

$$\mathrm{mIoU} = \frac{1}{K} \sum_{k=1}^{K} \mathrm{IoU}_k$$

where: TP_k is the number of true positives for class k, FP_k is the number of false positives, and FN_k is the number of false negatives.

Regarding the expected results, from the literature I expect that the dilated CNN may preserve more scene structure than a simple CNN. That is, a higher mIoU.

Expected Challenges

I expect several challenges during this project.

- Semantic segmentation is more difficult than image classification because every pixel must be labeled. Hence, computation would be relatively more expensive.
- The Cityscapes images are large, so training can become computationally expensive.
- A simple model may miss small objects or fine boundaries.
- Even with dilation, CNNs still have limitations in capturing long-range scene context (e.g., a photo of a bridge stretching from the far left edge to the far right edge of the frame). Does a dilated CNN improve this?

These challenges are part of why I find the project interesting. I want to better understand the tradeoff between model simplicity, spatial detail, and computation. The beauty of this work is that every result can be represented visually, and the performance can be judged accordingly.

References

[1] Fisher Yu and Vladlen Koltun. Multi-scale context aggregation by dilated convolutions. *International Conference on Learning Representations (ICLR)*, 2016.

[2] Liang-Chieh Chen, George Papandreou, Iasonas Kokkinos, Kevin Murphy, and Alan L. Yuille. Semantic image segmentation with deep convolutional nets and fully connected crfs. *International Conference on Learning Representations (ICLR)*, 2015.

[3] Jinglong Du, Lulu Wang, Yulu Liu, Zexun Zhou, Zhongshi He, and Yuanyuan Jia. Brain mri super-resolution using 3d dilated convolutional encoder–decoder network. *IEEE Access*, PP:1–1, 01 2020.

[4] Jonathan Long, Evan Shelhamer, and Trevor Darrell. Fully convolutional networks for semantic segmentation. *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 3431–3440, 2015.